

Process

How the recommendation was built, checked, and where it's still uncertain

1. How we built the agent

We used Claude Code with a Google Drive connector to build the model. It read the case PDF (Exhibit 4) and the provided Star River template, and we told it to build on that template, tie every input back to a case exhibit, and never hard-code a number, so the model stays correct if the inputs change.

The first attempt failed: Claude said it couldn't read the legacy `.xls` files and filled the model with invented numbers instead of flagging the problem. It turned out it could read the files fine — it just couldn't write back into them — so we had it work from copies instead, and told it to never invent data and to ask when something was missing.

Separately, Johnny Holloway built a verification layer on one rule: never verify a model by reading the model. It's two scripts plus an agent definition. `verification.py` rebuilds both scenarios independently from Exhibit 4 — every input typed from the case with the source sentence quoted beside it (`python verification.py --sources` prints them) — before the workbook is ever opened, then reconciles cell by cell. `audit_workbook.py` asks a different question: is the model built the way it claims to be? It walks every cell for typed numbers and searches the workbook for every figure the memo asserts. `.claude/agents/figure-checker.md` defines an agent that has to independently reproduce one claimed figure before calling it verified, and to report what it tried when it can't.

The first version of that verification script had its own bug: it applied an unquantified benefit to every year past the replacement date instead of just the three years before it, and returned SGD53,293 — matching neither the memo nor the correct figure. Perturbation testing (injecting a benefit and watching which years the gap between scenarios moved) caught it. The answer is years 1–3 only; that's now asserted in the script rather than assumed.

2. Division of labor

- Aidan Kunert** — built the Excel model (`.xlsx`) on the provided template, and the first draft of the memo.
- Johnny Holloway** — built the independent verification layer: `verification.py` , `audit_workbook.py` , `whatif.py` , the `figure-checker` agent, and findings 1–3 below.
- Matt Anderson** — received the process details from the rest of the team and compiled this file, `Process.md`.
- Britten Hager** — reviewed the whole process end to end and compiled the memo.
- (Per-commit attribution isn't captured in this note — pull it from the repo log if the submission needs it.)

3. How we checked the output

Not "we reviewed it." Each check below names the figure tested and what it was tested against.

Base case, reconciled to the cent. `verification.py` independently rebuilds all 28 year-by-year free cash flows and the six headline figures from Exhibit 4 before it ever reads the workbook:

	Independent check	The model
PV of costs, wait	(1,702,230.14)	(1,702,230.14)
PV of costs, buy now	(1,875,494.72)	(1,875,494.72)
Advantage of waiting	173,264.58	173,264.58
Advantage per year	25,669.32	25,669.32

This isn't circular: the check is built from the case PDF, not the workbook; it only reads the workbook's cached values, never its formulas, and only after computing its own; the reconciliation is year by year, so a structural error

(depreciation starting a year early, the maintenance contract escalating at the wrong rate) would show up in specific years rather than a uniform offset — all 28 match; and the check disagrees with the memo in three places (findings 1–3 below), which an echo of its own source couldn't do.

We also checked two figures directly against the case's own numbers: derived depreciation of 182,000 matches Exhibit 4's 182,000, and derived purchase-price savings of 286,877.50 matches Exhibit 4's 286,878. Both are live check cells in the workbook.

Structural audit. The Inputs sheet claims the Project Analysis sheet has no typed numbers, only formulas — verified across all 60 rows; the only literals are a zero on the year axis (D9) and a 0.01 rounding tolerance in a cross-check. All 13 inputs traceable to Exhibit 4 match the exhibit. The same audit searched the workbook for every figure the memo's Exhibit B claims and found none of the twelve sensitivity rows there (see Finding 2 below).

Horizon. Extending the 13-year horizon to 2015–2017 and recomputing independently gives a difference of exactly zero in every year — no longer horizon changes the answer. Truncating to 10 years understates the advantage of waiting by SGD44,425.

Sensitivities. Ten of the memo's twelve Exhibit B rows reproduce within SGD0.50, as do both break-even rates (6.61% cost of capital, 9.42% equipment escalation). The other two are Findings 1 and 3 (below and in section 5).

Scenario perturbation. We forced 2004 to maintenance 16,000, labor 60,000, overtime 20,000 and depreciation 70,000 to check the model flowed correctly: NPV moved by exactly the discounted cash-flow change, and the buy-now NPV didn't move at all, confirming nothing leaks between the two scenarios. We also reached the same NPV (173,264.58) by two independent routes and the same annualized figure (25,669.32) by two independent annualization methods.

4. Two things the agent got wrong

Finding 1 — the SGD34,000 break-even is understated by about 2.8x. The memo's risk section says unquantified benefits worth "more than about SGD34,000 a year, pre-tax, in perpetuity" would overturn the recommendation, and frames that as a low hurdle — "a little over 0.03% of sales." The correct figure is SGD93,910. The error: 34,000 is the equivalent annual advantage (25,669.32) divided by $(1 - 0.245)$, which grosses up a 13-year annualized figure as if the benefit gap lasted all 13 years. It doesn't — both scenarios run the new machine from 2005 on, so the benefit only differs in 2002–2004. Confirmed by perturbation: injecting a benefit of 1,000 moves the gap between scenarios only in years 1–3. (The memo's own labor break-even, SGD110,811, uses the correct three-year logic and reconciles exactly — so the memo contradicts itself on this point.) What we did about it: `verification.py` now computes the break-even by bisection instead of grossing up, and asserts the three-year property as a check rather than an assumption. The memo itself hasn't been updated to the corrected figure yet. The correction strengthens the recommendation — SGD93,910 a year is a much harder bar for the plant manager to clear than SGD34,000.

Finding 2 — right answer, unfindable reasoning. All twelve rows of the memo's Exhibit B, both break-even rates, and both break-even hurdles were searched for across every cell of the workbook — none of them is there. The workbook has no sensitivity machinery at all. Ten of the twelve numbers are nevertheless correct; we reproduced them independently. But as submitted, no one can point at a cell and say where any of them came from, and the assignment is explicit that every number in the memo should be traceable to the model. What we did about it: flagged this as a standing gap in `audit_workbook.py`. We haven't decided yet whether to add a sensitivity block to the workbook or to state plainly in the memo that Exhibit B was computed in `verification.py`, not the model.

5. What we are still unsure about

The "replace in 2003" sensitivity, SGD117,100, doesn't reproduce (Finding 3, unresolved). Running the model's own logic with replacement moved to 2003 gives SGD101,828, because the wait scenario's early-replacement path drops the old machine's remaining book value with no writeoff and no tax shield; crediting that shield gives SGD116,304 — still SGD796 short. We tried five different treatments of the old machine's writeoff and none lands on 117,100. The neighboring "replace in 2002" row reconciles exactly, at SGD58,455, but only with a writeoff the model doesn't implement elsewhere — so the two rows look like they were computed on different bases, and at least one wasn't computed in the workbook at all. (The buy-now scenario does credit the old machine's writeoff, SGD53,508 at $t=0$ — only the wait scenario's early-replacement path omits it.) This doesn't change the recommendation: waiting wins under every treatment we tried, by SGD26,318 to SGD117,100. What we can't

currently do is defend the specific number printed in the memo.

We used 2005 as the depreciation start year for the deferred machine's replacement. The case doesn't settle this, and 2004 is an equally defensible reading.

Writing off the old machine's full SGD218,400 at time zero is an inference — the template has a writeoff row, but the case doesn't spell out the timing.

We never tested the given 11% WACC against the comparables in Exhibit 5. It was handed to us as an assumption, not derived, so we used it as given.

The equivalent annual cost figures divide a present value that includes a $t=0$ cash flow by a 13-period ordinary annuity factor. Both alternatives are treated identically, so the comparison holds, and the NPV (row 50) is what actually decides the case — but the per-year figures are a restatement, not the decision rule, and the memo currently leads with them as if they were.